

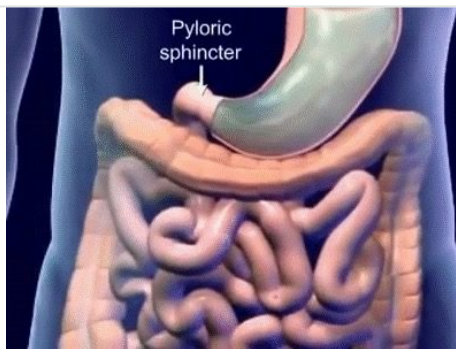
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Name the 3 regions of the small intestine, in order, with their relative lengths.
2. Which monosaccharide is absorbed by facilitated diffusion instead of sodium co-transport?
3. Which vitamin needs intrinsic factor, and in which region is it specifically absorbed?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Regions of the Small Intestine — the key discriminator

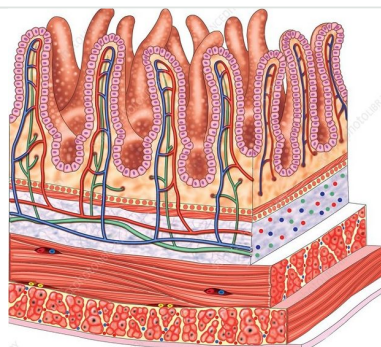
Region	Length	Distinguishing feature	Main role
Duodenum	~25 cm — shortest	Curves around head of pancreas; Brunner's glands secrete alkaline mucus; secretes secretin & CCK	Receives chyme + bile + pancreatic secretions; neutralizes acid
Jejunum	~2/5 of length	Richly vascular; prominent circular folds (plicae circulares); abundant villi	Main site of carbohydrate, protein, and lipid absorption
Ileum	~3/5 of length — longest	Fewer folds/villi than jejunum; contains Peyer's patches	Uptake of bile salts and vitamin B12 ; immune surveillance



Duodenum → jejunum → ileum, in continuity with the stomach.

2 Structural adaptations for absorption

The wall has 4 layers — mucosa, submucosa, muscularis, serosa. Circular folds, villi, and microvilli together massively increase the absorptive surface area. Each villus carries a capillary network **and** a lacteal, so absorbed nutrients have two possible routes into circulation.



Villus structure: capillary network and central lacteal within the lamina propria.

3 Motility patterns

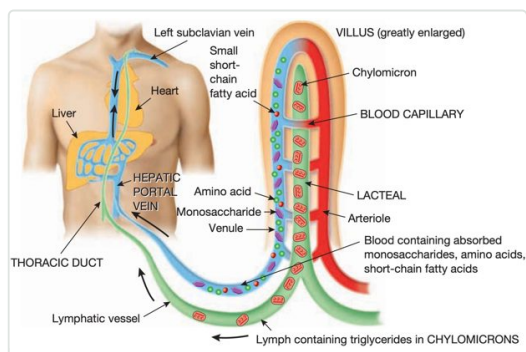
Pattern	Mechanism	Purpose
Segmentation	Local contractions	Mix chyme with digestive secretions
Peristalsis	Waves of contraction	Move chyme forward, maximise digestion & absorption

4 Brush border enzymes

NEVER MISS: brush border enzymes (lactase, maltase, sucrase, peptidases) are **bound to the microvilli membranes**, not secreted into the lumen — that's what separates them from pancreatic enzymes. They perform the **final** hydrolysis: disaccharides → monosaccharides, peptides → amino acids.

5 Digestion & absorption, macronutrient by macronutrient

Carbohydrates	Digestion: salivary/pancreatic amylase → polysaccharides to disaccharides → brush border enzymes → monosaccharides. Absorption: glucose & galactose via sodium-dependent co-transport; fructose via facilitated diffusion — the odd one out. All monosaccharides enter the bloodstream via the hepatic portal vein.
Proteins	Digestion: pepsin (stomach) → trypsin, chymotrypsin, carboxypeptidase (pancreatic) → brush border peptidases → amino acids. Absorption: amino acids via sodium-dependent active transport; di-/tripeptides via the proton-coupled transporter PepT1, then hydrolyzed to amino acids inside the enterocyte. Both routes end in the portal blood.
Lipids	Digestion: bile salts emulsify fat into micelles; pancreatic lipase hydrolyzes triglycerides to monoglycerides + free fatty acids; phospholipase and cholesterol esterase handle the rest. Absorption: lipid products diffuse into enterocytes, are re-esterified to triglycerides, combine with cholesterol/phospholipids/proteins to form chylomicrons, and are secreted into the lacteals — lymph, not portal blood.



Transport mechanisms for each nutrient class across the absorptive epithelium. Monosaccharides and amino acids go to blood via the hepatic portal vein; chylomicrons go to lymph via the lacteal and thoracic duct.

6 Macronutrient absorption — the exam-ready grid

Nutrient	Transport mechanism	Destination
Glucose / galactose	Na ⁺ -dependent co-transport	Blood (portal vein)
Fructose	Facilitated diffusion	Blood (portal vein)
Amino acids	Na ⁺ -dependent active transport	Blood (portal vein)
Di-/tripeptides	H ⁺ -coupled (PepT1)	Blood, after intracellular hydrolysis to AAs
Short-chain fatty acids	Simple diffusion	Blood (portal vein)
Long-chain fatty acids / monoglycerides	Micelle-mediated diffusion → chylomicrons	Lymph (lacteal)

7 Absorption of vitamins & minerals

Nutrient	Site / mechanism
Fat-soluble vitamins (A,D,E,K)	Absorbed with lipids via micelles
Water-soluble vitamins	Diffusion or carrier-mediated transport
Vitamin B12	Requires intrinsic factor; absorbed specifically in the ileum
Calcium	Duodenum; vitamin D-dependent
Iron	Duodenum; enhanced by gastric acid
Magnesium, phosphate, zinc	Throughout the small intestine

8 Water & electrolytes

Ion	Mechanism
Na⁺	Coupled with nutrient absorption + dedicated channels
Cl⁻	Passive absorption or exchange with HCO ₃ ⁻
K⁺	Mainly diffusion

Overall, the small intestine absorbs roughly **90%** of the nutrients and water that pass through the digestive system.

P Must-remember pearls

- Lipids are the **one** macronutrient that enters the **lymph** (chylomicrons via lacteals) — everything else goes to the hepatic portal vein.
- Vitamin B12 needs intrinsic factor **and** is absorbed specifically in the **ileum** — nowhere else.
- Fructose is the odd one out among monosaccharides: facilitated diffusion, not sodium co-transport.
- Brush border enzymes are membrane-bound, not secreted — that's what separates them from pancreatic enzymes.
- The small intestine absorbs **~90%** of all nutrients and water passing through the digestive system.

Q Test yourself

1. A patient with ileal resection (e.g. for Crohn's disease) is most likely to develop deficiency of:

- A. Iron
- B. Vitamin B12
- C. Calcium
- D. Folate absorbed in the jejunum

2. Which nutrient is absorbed via the lymphatic system rather than directly into portal blood?

- A. Glucose
- B. Amino acids
- C. Long-chain fatty acids (as chylomicrons)
- D. Short-chain fatty acids

3. A patient with lactase deficiency has impaired digestion at which step?

- A. Pancreatic amylase action
- B. Brush border disaccharide hydrolysis
- C. Bile emulsification
- D. Gastric acid hydrolysis

4. Calcium absorption in the duodenum depends on which vitamin?

- A. Vitamin B12
- B. Vitamin C
- C. Vitamin D
- D. Vitamin K

5. Which small intestinal region has the most prominent circular folds and villi, making it the main site of macronutrient absorption?

- A. Duodenum
- B. Jejunum
- C. Ileum
- D. All three equally

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — **B** — B12 absorption requires an intact ileum plus intrinsic factor; ileal loss causes B12 deficiency.
- 2 — **C** — long-chain fatty acids and monoglycerides form chylomicrons that enter lacteals, not blood capillaries.
- 3 — **B** — lactase is a brush border enzyme; its absence blocks the final hydrolysis of lactose to monosaccharides.
- 4 — **C** — duodenal calcium absorption is vitamin D-dependent.
- 5 — **B** — the jejunum has the richest vasculature, most prominent plicae circulares, and abundant villi.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Physiology GIT Block · MEDucation by Mattin Majid · GIT SGL slides, 2025-2026
Study alongside the lecture, not instead of it.